Mahmud Abdul Aziz Rohan

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Career Objective

I am an aspiring engineer committed to tackling challenges in local and international energy sectors using my expertise in energy engineering. I view myself as a lifelong learner, with a keen interest in advancing my knowledge of science, technology, and innovative engineering solutions.

Education

Khulna University of Engineering & Technology, Khulna-9203, Bangladesh 2022 – 2026
B.Sc. in Energy Science and Engineering | CGPA 2.80/4.00
Coursework: Thermodynamics | Fluid Mechanics | Heat and Mass Transfer | Power Plant Engineering | Renewable Energy Engineering | IC Engines | HVAC&R | Engineering Drawing and Simulation | Electrical Fundamentals

Govt. Science College, Tejgaon, Dhaka-1215 2018 – 2021
Higher Secondary Certificate | GPA 5.00/5.00

Bright School and College, Donia, Dhaka-1236 2018
Secondary School Certificate | GPA 5.00/5.00

Professional Experience & Workshops

Industrial Attachment, Karnaphuli Power Ltd, Chattogram Sep 2024

- Gained hands-on experience operating steam and diesel power-plant components, mastering key subsystems.

Training on Occupational Safety, Health & Environment Management May 2024
Department of Energy Science and Engineering, Khulna University of Engineering & Technology

- Conducted hazard identification and applied risk-analysis methods to improve occupational safety management.

Research & Project Experience

Energy, Exergy, and Exergoeconomic Analysis of Vapor Compression Refrigeration Cycle Using Low GWP Refrigerants, Undergraduate Thesis

- Investigated the effects of condensation and evaporation temperatures on key thermodynamic and exergoeconomic parameters (e.g., COP, operating cost) for basic and two-stage vapor compression refrigeration cycles, achieving an average improvement of 32% in COP and 24% in exergetic efficiency across multiple refrigerants.

Automated Water Level Controlling Using ESP32 and Micropython, 3^{rd} Year 2^{nd} Term Project

- Designed a real-time automated water reservoir monitoring system using an ESP32 microcontroller and MicroPython, reducing manual oversight and optimizing pump runtime cycles.

Skills

Engineering Tools & Software: CAD (SolidWorks, FreeCAD), MATLAB, LaTeX, Linux Environment
Technical & Data Analysis: Engineering Computation (NumPy, SciPy, SymPy), Data Visualization (Matplotlib), Predictive Modeling (Python), Thermodynamic Analysis (Energy, Exergy, Exergoeconomic)

Extracurricular Activities & Achievements

- Achieved First Runner-up position in Poster Presentation segment at Energy Fest 1.0.
- Acquired management experience as Treasurer at Energy and Automation Club.
- Secured an average time of 18.89 s in 3x3x3 Rubik's Cube segment at KUET MSE Cube Rush 2025.

References

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